

1.

$$(a) P(A \cap B) = P(A)P(B) = 0.4 \times 0.3 = \mathbf{0.12}.$$

$$(b) P(A \cup B) = P(A) + P(B) - P(A \cap B) \\ = 0.4 + 0.3 - 0.12 = \mathbf{0.58}.$$

2.

$$P(A | B) = 0.7 \neq P(A) = 0.5.$$

A and B are not independent, because independence requires $P(A | B) = P(A)$ when $P(B) > 0$.

3.

$$P(A^c) = 1 - 0.6 = 0.4.$$

$$P(B) = P(B | A)P(A) + P(B | A^c)P(A^c) \\ = 0.5 \times 0.6 + 0.2 \times 0.4 = 0.38.$$

$$P(A | B) = P(B | A)P(A) / P(B) \\ = (0.5 \times 0.6) / 0.38 = 15/19 \approx \mathbf{0.78947}.$$

4.

Let D = having the disease and + = a positive test.

$$P(D^c) = 1 - 0.02 = 0.98.$$

$$P(+ | D^c) = 1 - 0.90 = 0.10.$$

$$P(+) = P(+ | D)P(D) + P(+ | D^c)P(D^c) \\ = 0.95 \times 0.02 + 0.10 \times 0.98 = 0.117.$$

$$P(D | +) = P(+ | D)P(D) / P(+) \\ = (0.95 \times 0.02) / 0.117 = 19/117 \approx \mathbf{0.16239 (16.24\%)}.$$

5.

$$(a) E[X] = \sum x \cdot P(X = x) \\ = 85(0.375) + 90(0.375) + 95(0.125) + 100(0.125) \\ = 31.875 + 33.750 + 11.875 + 12.500 = \mathbf{90}.$$

$$(b) E[X^2] = 85^2(0.375) + 90^2(0.375) + 95^2(0.125) + 100^2(0.125) \\ = 2709.375 + 3037.500 + 1128.125 + 1250.000 = 8125.$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 8125 - 90^2 = \mathbf{25}.$$

$$(c) \text{Sample mean} = (85 + 90 + 85 + 95 + 90 + 85 + 100 + 90) / 8 \\ = 720 / 8 = \mathbf{90}, \text{ which equals } E[X].$$

6.

$$(a) H(X) = -\sum p(x) \log_2 p(x) \\ = -[0.4 \log_2(0.4) + 0.3 \log_2(0.3) + 0.2 \log_2(0.2) + 0.1 \log_2(0.1)] \\ = 0.528771 + 0.521090 + 0.464386 + 0.332193 \\ \approx \mathbf{1.84644 \text{ bits}}.$$

(b) Each message has probability 1/4.

$$H(X) = -4 \times (1/4) \times \log_2(1/4) = -1 \times (-2) = \mathbf{2 \text{ bits}}.$$